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STOCHASTIC PROGRAMMING

PROBLEM SESSION 2

10. A computer manufacturer has to plan the next season production of 3 types of computers. Expected benefits for each model unit are b_i, $i=1_3$ currency units. The company will buy now all the components for building the computers, excluding the memory modules and hard drives: 2 memory modules and 1 hard drive for the first type; 2 memory modules and hard drives for second type; 1 memory modules and 2 hard drives for 3rd type. However, the available number of memory modules and hard drives in the market at the time of building the computers will be independent r.v. ξ_1 and ξ_2. For each required, but no available, memory module or hard drive the company estimates costs (benefit losses) of c_1 (for memory modules) and c_2 (for hard drives) currency units.

a) Formulate the stochastic opt. prob. to be solved by the company to get opt. prod. planing (# computers per type, max. benef. and reduce costs by lack of memory and hard drives)

x_i = # of computers type i

$$\max \sum_i b_i x_i - \mathbb{E}_\xi [Q(x,\xi)]$$
$$x_i \in \mathbb{Z}_+, \forall i$$

$$Q(x,\xi) = \min c_1 y_1 + c_2 y_2$$
$$2x_1 + 2x_2 + x_3 + y_1 \geq \xi_1$$
$$x_1 + 2x_2 + 2x_3 + y_2 \geq \xi_2$$
$$\underbrace{y_1, y_2}_{\text{\# of disks extra needed}} \in \mathbb{Z}_+$$

(b) Formulate a new problem by replacing 2nd stage by the following chance constraint: "the prob. that the required number of memory modules and hard drives is less or equal than ξ_1 and ξ_2, respc., has to be $\geq \alpha$". ξ_1 and ξ_2 are random. Simplify chance constraint to obtain a linear expression, assuming that variables $\xi_1 - 100$, $\xi_2 - 200$ have an exp. dist. of params. λ_1, λ_2. That is

$$F_{\xi_1}(x) = 1 - e^{-\lambda_1 (x-100)} \quad \text{and} \quad F_{\xi_2}(x) = 1 - e^{-\lambda_2 (x-200)}$$

$$\mathbb{P}(2x_1 + 2x_2 + x_3 \leq \xi_1,\ x_1 + 2x_2 + 2x_3 \leq \xi_2) \geq \alpha$$

$$\parallel$$

$$\underbrace{\mathbb{P}(2x_1 + 2x_2 + x_3 \leq \xi_1)} \cdot \underbrace{\mathbb{P}(x_1 + 2x_2 + 2x_3 \leq \xi_2)} \geq \alpha$$

$$\underbrace{\left[1 - F_{\xi_1}(2x_1 + 2x_2 + x_3)\right]} \cdot \left[1 - F_{\xi_2}(x_1 + 2x_2 + 2x_3)\right] \geq \alpha$$

$$e^{-\lambda_1 (2x_1 + 2x_2 + x_3 - 100)} \cdot e^{-\lambda_2 (x_1 + 2x_2 + 2x_3 - 200)} \geq \alpha$$

$$-\lambda (2x_1 + 2x_2 + x_3 - 100) - \lambda_2 (x_1 + 2x_2 + 2x_3 - 200) \geq \ln(\alpha)$$

Therefore

$$\max \quad b_1 x_1 + b_2 x_2 + b_3 x_3$$

$$(2\lambda_1 + \lambda_2) x_1 + (2\lambda_1 + 2\lambda_2) x_2 + (\lambda_1 + 2\lambda_3) x_3 \leq 100\lambda_1 + 200\lambda_2 - \ln(\alpha)$$

$$x_1, x_2, x_3 \in \mathbb{Z}_+$$

Stochastic Programming.
Problems session 2

7. An airline wants to decide the number of "business class", "first class" and "tourist class" seats (x_1, x_2 and x_3) to be offered in its aircrafts. In the following we will assume subscripts $_1$, $_2$ and $_3$ refer, respectively, to "business class", "first class" and "tourist class". Each "business class" and "first class" seats occupy the same space than 2 and 1.5 "tourist class" seats respectively. The maximum number of "tourist class" seats in the entire plane is N. In addition, for each group or fraction of n_1 n_2 or n_3 ($n_1 < n_2 < n_3$) seats offered (not purchased) the airline has to hire a flight attendant. The maximum number of flight attendants in the aircraft is H. Once fixed, the number of seats per class can not be modified. The demand for each type of seat are random variables $\xi_i, i = 1, \ldots, 3$. Earnings per seat sold for each class are $b_i, i = 1, \ldots, 3$ €.

   (a) Formulate the stochastic optimization problem to be solved by the company. The goal is to obtain the number of seats per class that maximize the earnings. Note that variables are integer.

   (b) Write the extensive form of the problem, assuming that $\xi_i, i = 1, \ldots, 3$ are totally correlated random variables with s values $\xi_{i_j}, i = 1, \ldots, 3, j = 1, \ldots, s$, of probabilities $p_j, j = 1, \ldots, s$.

8. (a) A company manufactures n products from m raw materials. Data associated to products and raw materials will be indexed, respectively, by $i = 1, \ldots, n$ and $j = 1, \ldots, m$. One unit of product i requires $0 \leq a_{ij} \in \mathbb{Z}$ units of raw material j. The purchasing unit price of raw material j is c_j€/unit. The selling unit benefit or product i is b_i€/unit. The demand for each product will be a random variable ξ_i with a certain distribution. The company acquires raw materials before knowing the demand. Each non-used unit of raw material j will have a residual value of r_j€/unit, but it will incur in an storage cost of w_j€/unit. Unmeet demand will have a high and fixed cost of p_i€ (due to loss of confidence for future orders); note **this is a fixed cost, not a unit cost**. Formulate the stochastic problem to be solved by this company to find out how many units of raw materials to purchase to maximize profits (of minimize costs).

   (b) Consider now that the company is concerned about satisfying all the demand. It wants to know the optimal number of units of raw material to acquire, with purchase costs c_j, such that the probability that all demands ξ_i are satisfied is greater than or equal to $1 - \alpha$, where $\alpha \in (0, 1)$ (in practice α is a small value). As before, a_{ij} denotes the units of raw material j needed for a unit of product i.

   - Formulate the problem with chance constraints to be solved by the company.
   - Solve the chance constrained problem for the particular case of $m = 2$ raw materials, $n = 1$ products, $(c_1, c_2) = (5, 4)$, $(a_{11}, a_{12}) = (1, 2)$, ξ_1 is a uniform random variable of values [0,200] (assume continuous, to simplify the solution), and $\alpha = 0.01$.

9. (a) A company has to install n pumps in a recently built gas station. Each pump will deliver one (and only one) of k different fuels (gasoline 95, gasoline 98, diesel, diesel-plus, etc). Each pump has an hourly cost associated to a pollution tax. These costs, denoted as $t_i, i = 1, \ldots, k$, depend on the particular fuel delivered (we assume this is a unit cost per pump, not per liters of fuel delivered). The demand (considered as number of hourly customers to the gas station) for each fuel is a known discrete random variable $\xi_i, i = 1, \ldots, k$. The company estimates that the number of hourly services in each pump is at most 20. If the average number of hourly costumers to each pump of fuel i (computed as ξ_i/x_i, where x_i is the number of pumps delivering fuel i) is greater than 20, then some potential services (those exceeding 20) will be missed (customers will depart without a service for excessive waiting time). The hourly benefit for each service of fuel $i = 1, \ldots, k$, is b_i.

   Formulate the stochastic programming problem to be solved by the company to obtain the optimal number of pumps for each fuel, that is, the one giving the maximum net benefit (benefit minus cost). (Note: you can not use "if-then-else" expressions, just formulate valid linear or integer optimization problems for the first and second stage).

(b) Consider a variant of the above problem with probabilistic constraints. The number of pumps is $n = 4$, and the number of fuels is $k = 2$. The company has a net benefit of 1 units per pump of fuel 1, and 1/2 units per pump of fuel 2. We will denote as $\xi_i, i = 1, 2$, the number of hourly costumers for fuel i. The company wants to compute the optimal number of pumps for fuel 1 and 2, such that the probability that ξ_i/x_i (average number of hourly costumers to each pump of fuel i) is greater than 20, has to be less or equal than some small α, for each fuel (that is, the problem has two probabilistic constraints, one for each fuel). Assuming, for simplicity, that ξ_1, ξ_2 have continuous distributions $\xi_1 \sim U[0, 40]$ and $\xi_2 \sim U[0, 20]$, and $\alpha = 0.1$, formulate and solve the resulting problem with probabilistic constraints.

10. A computer manufacturer has to plan the next season production of 3 types of computers. Expected benefits for each model unit are $b_i, i = 1, \ldots, 3$ currency units. The company will buy now all the components for building the computers, excluding the memory modules and hard drives, which be purchased in the future at the time of building the computers, since it is expected their prices will be lower. Each type of computer requires a number of memory modules and hard drives: 2 memory modules and 1 hard drive for the first type; 2 memory modules and 2 hard drives for the second type; and 1 memory modules and 2 hard drives for the third type. However, the available number of memory modules and hard drives in the market at the time of building the computers will be independent random variables ξ_1 and ξ_2. For each required, but no available, memory module or hard drive the company estimates costs (benefit losses) of c_1 (for memory modules) and c_2 (for hard drives) currency units.

    (a) Formulate the stochastic optimization problem to be solved by the company to get the optimal production planning (number of computers per type, to maximize benefits and to reduce the costs by lack of memory modules and hard drives).

    (b) Formulate a new problem by replacing the second stage by the following chance constraint: "the probability that the required number of memory modules and hard drives is less or equal than ξ_1 and ξ_2, respectively, has to be greater than or equal to α". Take into account that ξ_1 and ξ_2 are independent. Simplify the chance constraint to obtain a linear expression, assuming that variables $\xi_1 - 100$ and $\xi_2 - 200$ have an exponential distribution of parameters λ_1 and λ_2, that is, the probability distributions of ξ_1 and ξ_2 are $F_{\xi_1}(x) = 1 - e^{-\lambda_1(x-100)}$ and $F_{\xi_2}(x) = 1 - e^{-\lambda_2(x-200)}$.